Part IV The Kähler case

Line bundles & projective embeddings

<u>Defn</u> A holo. line bundle L over a complex mfld M is a complex mfld with projection map $\pi : L \to M$ s.t.

1) $\exists$ cover $M = \bigcup_\alpha U_\alpha$ and bihol. $\varphi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{C}$

$$\begin{tikzcd} \pi^{-1}(U_\alpha) \arrow[rr, "\varphi_\alpha"] \arrow[dr, "\pi"'] & & U_\alpha \times \mathbb{C} \arrow[dl, "pr_1"] \\ & U_\alpha & \end{tikzcd}$$

2) transition maps

$$\varphi_{\alpha\beta} : \varphi_\alpha \circ \varphi_\beta^{-1} : (U_\alpha \cap U_\beta) \times \mathbb{C} \to (U_\alpha \cap U_\beta) \times \mathbb{C}$$

are complex linear in $\mathbb{C}$ factor.

$\Gamma(M, L)$ = smooth sections of L
$= \{ s : M \to L \text{ smooth} \mid \pi \circ s = id_M \}$

$H^0(M, L)$ = holomorphic sections of L

<u>Thm</u> If M is compact then $\dim H^0(M, L) < \infty$.

E.g. 1) $\mathcal{O}_{\mathbb{P}^n}(-1) \subset \mathbb{CP}^n \times \mathbb{C}^{n+1}$

$$= \left\{ ([z], w) \;\middle|\; \begin{array}{l} z \in \mathbb{C}^{n+1} \setminus \{0\} \\ w = \lambda z \text{ for some } \lambda \in \mathbb{C} \end{array} \right\}$$

$$\mathcal{O}_{\mathbb{P}^n}(1) = \mathcal{O}_{\mathbb{P}^n}(-1)^* \quad (\text{dual})$$

$$\mathcal{O}_{\mathbb{P}^n}(k) = \mathcal{O}_{\mathbb{P}^n}(1) \otimes \dots \otimes \mathcal{O}_{\mathbb{P}^n}(1) \quad (k \text{ factors})$$

$$H^0(\mathbb{CP}^n, \mathcal{O}_{\mathbb{P}^n}(k)) = \text{homog. deg } k \text{ poly. on } \mathbb{C}^{n+1}$$

2) If $M \subset \mathbb{CP}^N$ projective mfld,

$$L = \mathcal{O}_{\mathbb{P}^N}(k)\big|_M$$

$$H^0(M, L) = \left\{ \text{deg. } k \text{ poly. on } \mathbb{C}^{N+1} \right\} \Big/ I_k$$

$\uparrow$ poly. vanishing on M

<u>Maps to $\mathbb{CP}^N$</u>

Let $S = \{ s_0, \dots, s_N \} \subset H^0(M, L)$

These define holo. map

$$\Phi_S : M \setminus B \longrightarrow \mathbb{CP}^N$$
$$x \longmapsto [s_0(x) : \dots : s_N(x)]$$

where B = base locus $= \{ x \in M \mid s_i(x) = 0 \ \forall i \}$

Usually we take S to be a basis of $H^0(M, L)$ and write Φ_L.

Metrics and curvature

A Hermitian metric on L is a Hermitian inner product $\langle \cdot, \cdot \rangle$ on each fiber $\pi^{-1}(x)$ varying smoothly.

Chern connection: $\exists$ unique connection

$$\nabla : \Gamma(M, L) \longrightarrow \Gamma(M, L \otimes \Omega^1 M)$$

compatible with
- the Hermitian metric:
$$v\langle s, t\rangle = \langle \nabla_v s, t\rangle + \langle s, \nabla_v t\rangle$$
- the holo. structure:
$$\nabla^{0,1} = \overline{\partial}_L$$
i.e. $\nabla^{0,1} s = 0$ if s is holomorphic

can extend ∇ to
$$\nabla : \Gamma(M, L \otimes \Omega^1(M)) \longrightarrow \Gamma(M, L \otimes \Omega^2(M))$$
by $\nabla(s\alpha) = \nabla s \wedge \alpha + s\, d\alpha$

<u>Curvature</u> of ∇ is 2-form Ω s.t.
$$\nabla^2 s = s\Omega$$

<u>Locally</u>: Let s be a non-vanishing holo. section of L over $U \subset M$.

Let $h(s) = |s|^2$, $\nabla s = s\alpha$, $\alpha \in \Omega^{1,0} M$

$$\partial h = \partial\langle s, \bar{s}\rangle = \langle \nabla^{1,0} s, \bar{s}\rangle + \cancel{\langle s, \nabla^{1,0} \bar{s}\rangle}$$
$$= \alpha h$$

$$\alpha = h^{-1}\partial h = \partial \ln h$$

$$\nabla^2 s = (\nabla s)\wedge\alpha + s\, d\alpha$$
$$= s\, \bar\partial\partial \ln h$$

i.e. curvature is $\boxed{\Omega = \bar\partial\partial \ln h.}$

<u>Defn</u> (L,h) is positive, if $\sqrt{-1}\,\Omega$ is a Kähler form.

<u>E.g.</u> 1) $L_{\mathbb{C}} = \mathbb{C}^n \times \mathbb{C}$ with metric h defined by $h(1) = e^{-|z|^2}$ has curvature

$$\Omega = \bar\partial\partial(-|z|^2) = \partial\bar\partial |z|^2$$

$\sqrt{-1}\,\Omega = \sqrt{-1}\,\partial\bar\partial|z|^2 =$ standard Kähler form on $\mathbb{C}^n$.

2) On $\mathcal{O}_{\mathbb{P}^n}(-1)$ natural $U(n+1)$-invariant metric has curvature

$$\Omega = \bar\partial\partial \ln\|s\|^2 \quad \text{where}$$

$s : U \longrightarrow \mathbb{C}^{n+1}$ is a holo. section, $U \subset \mathbb{CP}^n$

i.e. $\omega_{FS} = -\sqrt{-1}\,\Omega$.

<u>Thm</u> (Kodaira embedding)

Let $(L,h) \to M$ be a positive Hermitian line bundle, over compact M. Then for sufficiently large k the maps

$$\Phi_{L^k} : M \longrightarrow \mathbb{CP}^{N_k}$$

$(L^k = L \otimes \cdots \otimes L)$

are embeddings.

<u>Thm</u> (Donaldson - Sun)

Given n, D, v, $\exists k = k(n, D, v) > 0$ s.t. if $(L,h) \to M$, $\omega = \sqrt{-1}\Omega > 0$, over compact M satisfies

1) $|Ric(\omega)| \leq 1$
2) $Vol(M,\omega) > v$
3) $Diam(M,\omega) < D$

Then $\Phi_{L^k} : M \longrightarrow \mathbb{CP}^N$ is an embedding.

(Liu - Sz. in 1), $Ric(\omega) > -\omega$ is enough)

The proof gives more: if $(L_i, h_i) \to M_i$ sequence as above and $(M_i, \omega_i) \xrightarrow{GH} (Z, d_Z)$ then Z is homeomorphic to normal projective variety.